

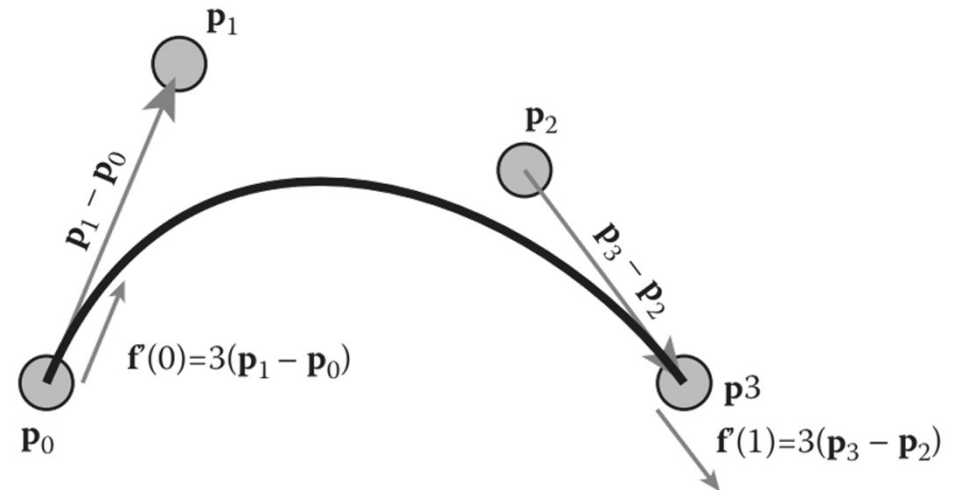
Lecture 8 — Curve in Computer Graphics

FreeCAD 3D Modeling Training
Engineering and Computer Graphics

Focus: Hermite, Cardinal, Bézier, B-Spline, NURBS —
and how they appear in OpenGL and FreeCAD

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Theory 45 min | Guided practice 90 min



Learning goals and lecture structure

Outcomes

- Describe Hermite, Cardinal, Bézier, B-Spline and NURBS curves.
- Explain continuity, local control, knots and weights.
- Relate matrix-based curve evaluation in OpenGL to curve tools in FreeCAD.

Timing

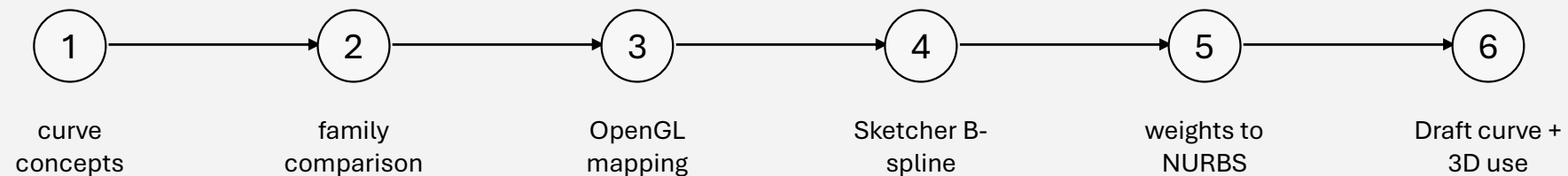


45 min theory
90 min step-by-step
practice

What students already know

- OpenGL matrices, lighting, textures, blending.
- Basic FreeCAD sketching, pad and pocket operations.
- Laptop-based classroom workflow.

Today's sequence



One mathematical idea, two toolchains

A curve is an infinitely large set of points

Curve mathematics

- A curve is a parametric mapping from parameter u to position:
 $(x,y) = \mathbf{f}(u)$
- Continuity tells how smoothly pieces meet.
 C^n continuity
- Control points, tangents, knots and weights change behavior.

OpenGL mental model

- Sample u to generate vertices or tessellated fragments.
- Use basis matrices / blending functions to evaluate position and derivatives.
- Continuity affects shading, normals, motion and sweep paths.

FreeCAD mental model

- Sketcher and Draft expose control points, degree, knots and weights.
- B-spline tools show control polygon and curvature comb.
- Curve choice affects editability and the quality of later solids/surfaces.

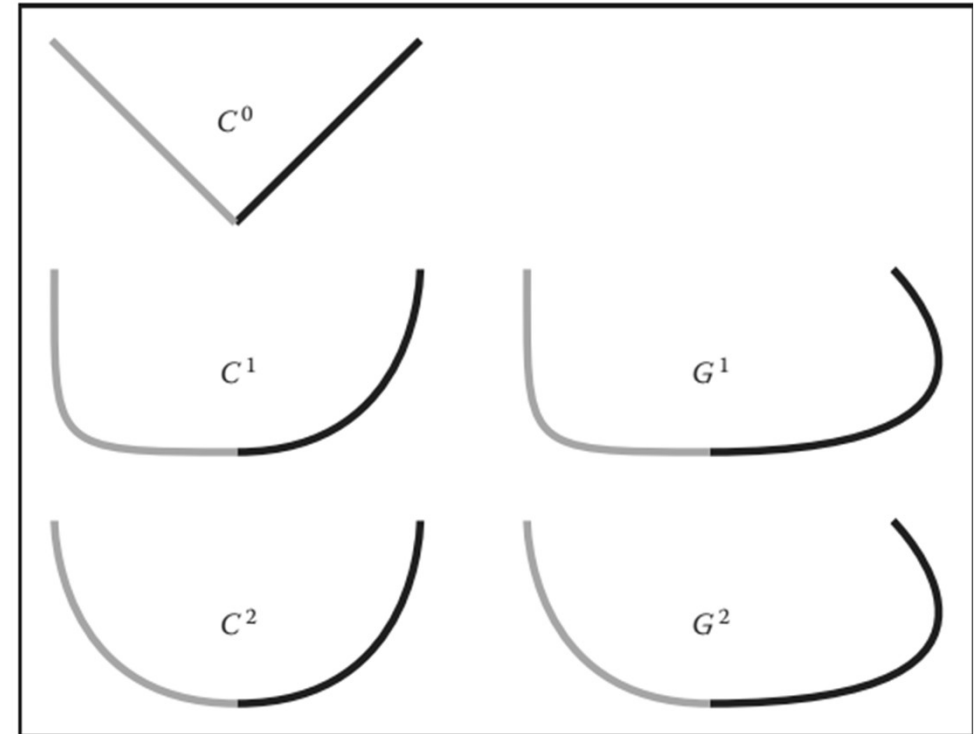
“Local control matters in both rendering and CAD editing.”

Curve essentials: parameter, continuity, and local control

Key ideas to watch throughout the lecture

- Parametric form: move a point along the curve by changing u : $f(u)=(x,y)$
- C^0 means connected, C^1 adds tangent continuity, C^2 adds curvature continuity.
- Geometric continuity (G^1 , G^2) can match shape even when parameter speed differs.
- **Local control** means one edit changes only a nearby region; this is crucial in CAD.

Visual cue: a sharp corner usually means C^0 but not C^1 .



C^n continuity

$$\frac{d^n f_1(u)}{du^n} = \frac{d^n f_2(u)}{du^n}$$

G^n continuity

$$\frac{d^n f_1(u)}{du^n} = k \frac{d^n f_2(u)}{du^n}$$

Curve families I — Hermite and Cardinal

Hermite

endpoints + tangents

Cardinal

interpolating cubic, tension

Bézier

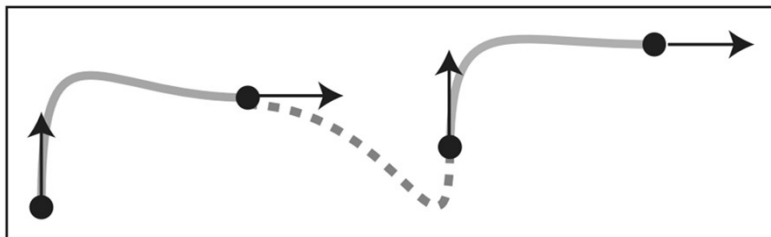
control polygon, convex hull

B-Spline / NURBS

local control, knots, weights

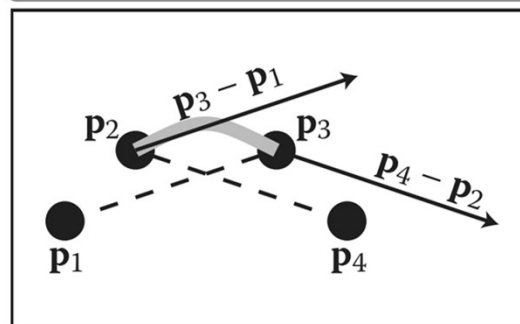
Hermite

- Specify (**position + first derivative**) at the **start** and **end**.
- Natural for animation paths or when endpoint tangents are known.
- Good conceptual bridge from vectors in OpenGL to editable tangents in CAD.



Cardinal

- Interpolates **points** with **C1 continuity + local control**.
- Uses neighboring points to infer tangents.
- Tension parameter (t) changes how tightly the curve bends;
- Catmull–Rom is the tension = 0 special case.



$$\begin{aligned} \mathbf{f}'(0) &= \frac{1}{2}(1-t)(\mathbf{p}_3 - \mathbf{p}_1), \\ \mathbf{f}'(1) &= \frac{1}{2}(1+t)(\mathbf{p}_4 - \mathbf{p}_2). \end{aligned}$$

Curve families II — Bézier

Progressive disclosure: one family emphasized at a time

Hermite

endpoints + tangents

Cardinal

interpolating cubic, tension

Bézier

control polygon, convex hull

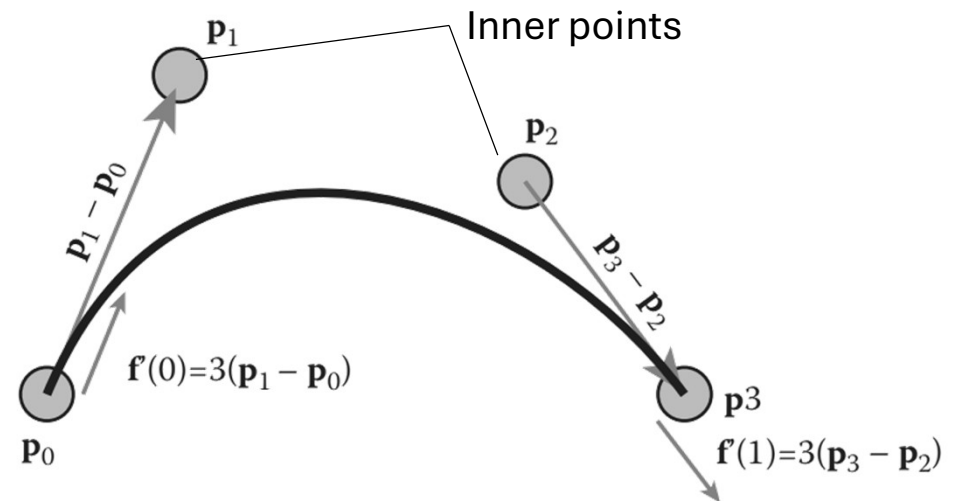
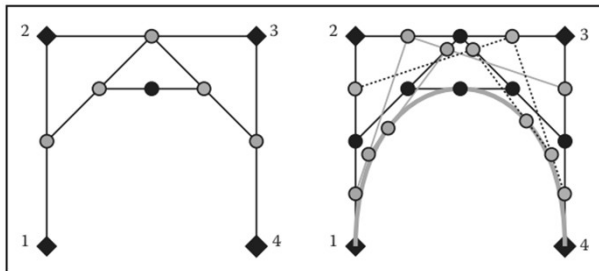
B-Spline / NURBS

local control, knots, weights

Bézier curves

- Interpolates first and last control points.
- Inner points pull the shape but are not usually interpolated.
- Convex-hull and variation-diminishing properties make behavior predictable.
- de Casteljau gives a stable evaluation / subdivision algorithm.

de Casteljau
algorithm for a
cubic Bézier



$$\mathbf{p}(u) = \sum_{k=0}^n p_k C(n, k) u^k (1-u)^{(n-k)}.$$

Curve families III — B-Spline and NURBS

Progressive disclosure: one family emphasized at a time

Hermite

endpoints + tangents

Cardinal

interpolating cubic, tension

Bézier

control polygon, convex hull

**B-Spline /
NURBS**

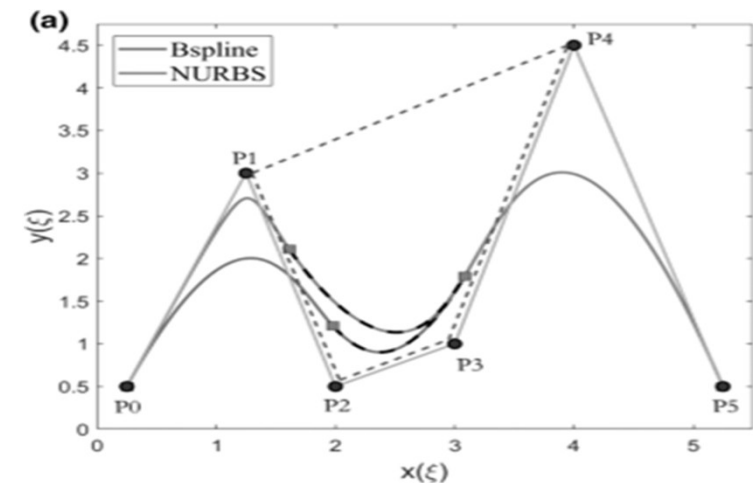
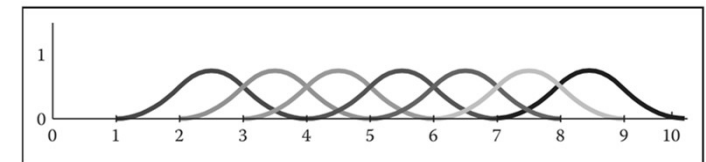
local control, knots, weights

B-Spline / NURBS focus

- B-splines are piecewise polynomial and usually approximate control points.
- **Local control:** each point on the curve depends on only a limited neighborhood.
- Knots and knot multiplicity change continuity and interpolation behavior.
- NURBS add weights, enabling exact conic sections and more shape control.

$$P(u) = \sum_{i=0}^n N_{i,k}(u) P_i,$$

$N_{i,k}(u)$ is non-zero
at small range of u



B-Spline Vs. NURBS

Best use of each tool

Feature	Bézier Curve	B-Spline	Hermite Spline	Cardinal Spline
Passes Through Points?	Approximating: Only starts/ends at points.	Approximating: Usually misses all points.	Interpolating: Passes through every point.	Interpolating: Passes through every point.
Control Input	Control Points (Magnets).	Control Points + Knot Vector.	Points + Tangent Vectors (Angles).	Points + Tension Parameter.
Local vs. Global	Global: Move one point, the whole curve moves.	Local: Moving a point only affects its neighborhood.	Local: Tangents only affect the immediate segment.	Local: Tension and points affect local shape.
Degree Control	Tied: Degree increases with the number of points.	Independent: You choose the degree (usually cubic).	Fixed: Almost always Cubic (degree 3).	Fixed: Almost always Cubic (degree 3).
Best Use Case	Graphic design, logos, and font outlines.	CAD, 3D modeling, and high-end engineering.	Keyframe animation (controlling velocity/angle).	Game paths, waypoints, and data visualization.

OpenGL and FreeCAD mapping for curves

What about PowerPoint?

Concept	In OpenGL thinking	In FreeCAD thinking
Parameter u	Sample positions on the curve for tessellation or drawing	Drag or inspect the curve while FreeCAD evaluates positions internally
Control points	Input data in a buffer / uniform structure	Editable points, tangents, knots, handles
Basis / blending	Hermite, Bézier or B-spline basis matrices	Sketcher and Draft tools hide the math but expose the result
Continuity	Affects smooth normals, motion and sampling quality	Affects visual fairness and the quality of later sweeps/lofts
Weights / knots	Extra parameters change interpolation and local influence	NURBS weights and knot multiplicity are editable design controls

Teaching tip: answer “why continuity and knots matter” with the downstream effect on editing locality and 3D surface quality.

Practice roadmap: 5 activities, one shared model

A

Sketcher B-spline

Create a curve from control points, inspect degree, and talk about continuity.

B

Weights → rational behavior

Adjust weights and see why a rational B-spline behaves differently from a plain polynomial spline.

C

Draft curve → 3D use

Compare Bézier and B-spline editing, then reuse a curve as profile or path for later modeling.

D

Turn a Draft curve into a 3D feature

E

Use Shape From Text for raised or engraved features

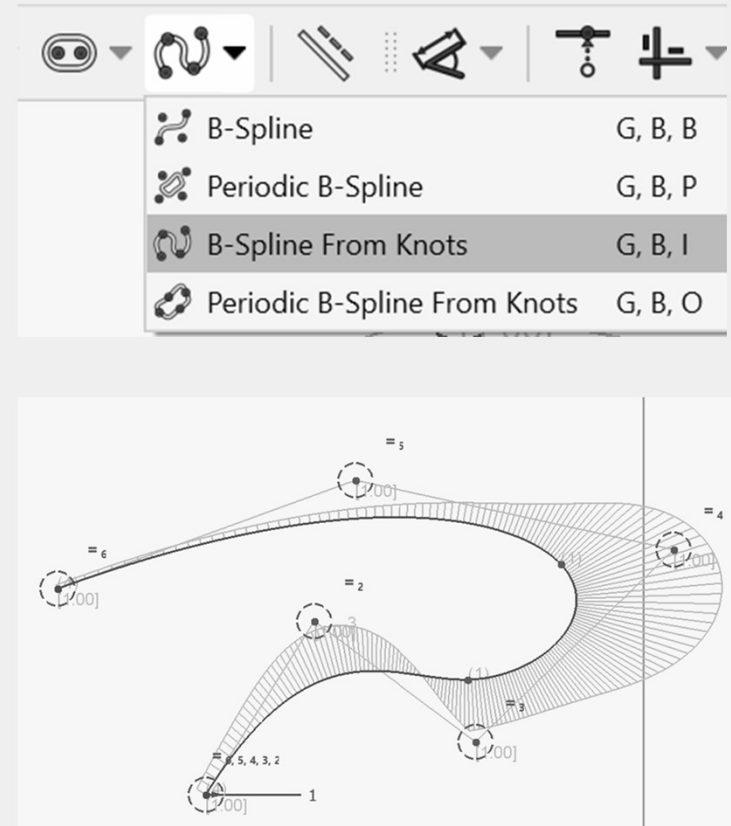
Practice A — Create a B-spline in Sketcher

Workflow

- Create a new sketch on the XY plane.
- Choose the Sketcher B-Spline tool (
- Left-click several control points; right-click to finish.
- Press U/J while drawing if you want to change the degree explicitly.
- Try to fix most of the points and move one

Checkpoint prompt

Does your curve interpolate every control point? If not, why is that still useful?



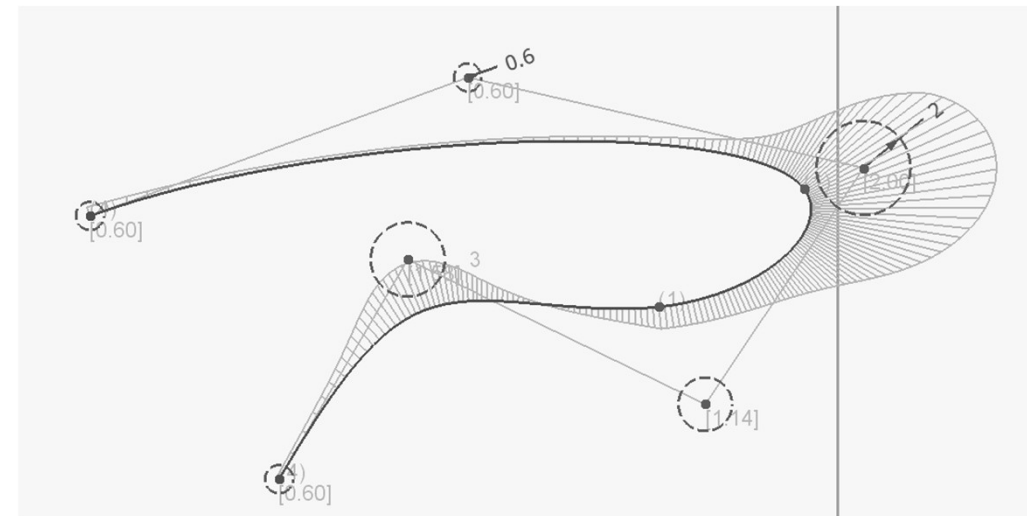
Practice B — Weights, knots, and the jump to NURBS

What students do

- Display the weight circles around control points.
- Break the equality of the radii so weights can vary independently.
- Increase one weight and observe the curve being attracted to that control point. (select circle + D)
- Discuss why zero and negative weights are not allowed in the usual workflow.
- Relate nonuniform knots or repeated knots to interpolation behavior.

Observe and explain

A rational B-spline is a NURBS curve. The “R” comes from the ratio introduced by the weights.



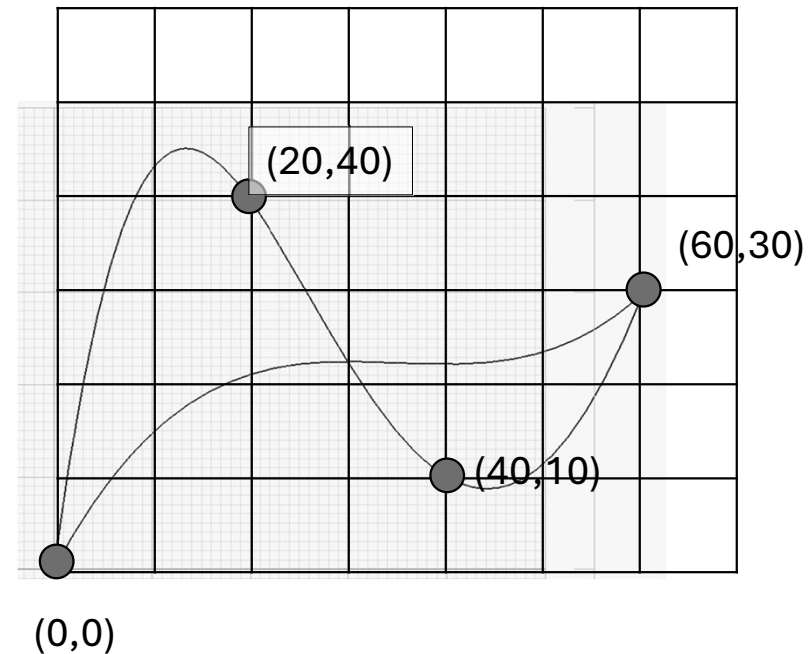
Practice C — Compare Draft Bézier and Draft B-spline

Compare editing behavior

- Create a Draft Bézier curve from points.
- Create a Draft B-spline from a similar point set.
- Move one interior control point in each case.
- Ask: which edit feels more local? which curve is easier to shape globally?

Then reuse the result

- Use the curve as a path or profile in a later Sweep / Loft exercise.
- Connect back to the students' prior pad / pocket experience.



Remember to unselect relative coordinate

☐ Relative (R)

Why this matters for modeling

The curve you sketch is not just 2D decoration. Its continuity and locality shape the surfaces and solids you create next.

Practice D — Turn a Draft curve into a 3D feature

Workflow

Create a Draft Wire, B-spline, or Bézier that forms a closed profile.

Use Draft Edit to move points until the profile is clean and closed.

For a quick solid: switch to Part Workbench and use Extrude.

For PartDesign editing later: use Draft to Sketch, activate a Body, then Pad or Pocket.

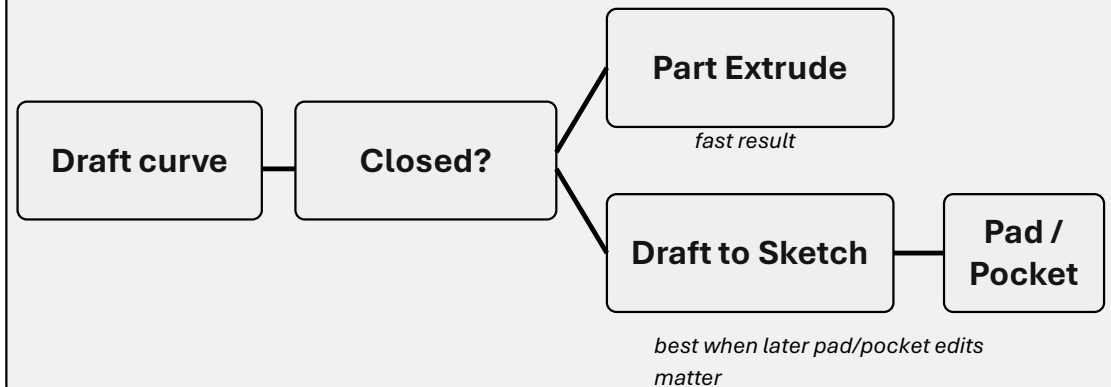
If the curve is open, keep it for sweep/path work instead of pad/pocket.

Checkpoint prompt

Which workflow preserves the most editability for later constraints?

Classroom demo script

1. Draw one simple closed outline from curve tools in Draft.
2. Show the same outline in two branches: Part Extrude vs Draft to Sketch.
3. Edit one control point and ask which branch is easier to revise.
4. Reinforce the rule: closed profile for solid features, open curve for paths.



Practice E — Use Shape From Text for raised or engraved features

ShapeString workflow

- 1) Switch to Draft Workbench and launch Shape From Text.
- 2) Set the text string, choose a font file, then define size and placement.
- 3) Place the text on a plane or on the face that will receive the feature.
- 4) For raised text: use PartDesign Pad or Part Extrude.
- 5) For engraved text: use PartDesign Pocket to cut the letters into the solid.

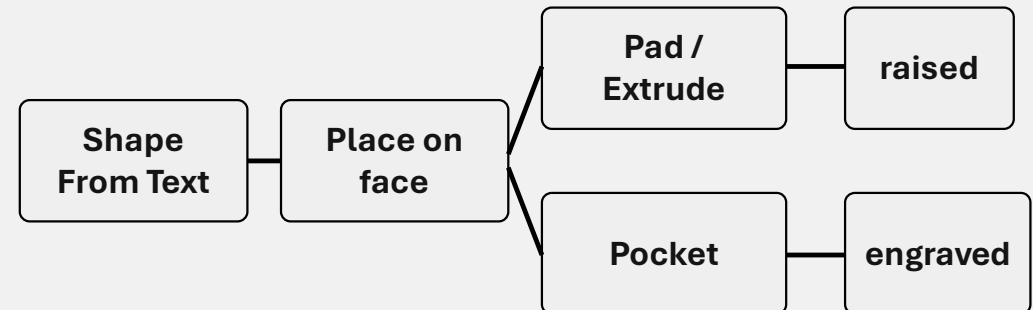
Common pitfall

The letters are separate closed loops; keep them aligned with the support face before you cut.

Extruding a ShapeString produces a compound of solids, so selection order and placement matter.

What students should watch

- Text behaves like geometry, not editable typography.
- The font file controls the letter outlines.
- Closed inner loops (A, O, P, R...) become holes in the final feature.
- Pocket cuts inward; Pad or Extrude adds material outward.



Mini challenge: engrave your initials on a small block, then change the font and compare edge quality.

Decision rule — which Draft output should become Extrude, Pad, or Pocket?

Starting object	Best next step	Why	Do this in class
Closed Draft curve	Part Extrude	Fast 3D result from a profile you already drew.	Use when speed matters more than Body history.
Closed Draft curve needing later feature edits	Draft to Sketch → Pad/Pocket	Best route when the profile must stay editable in PartDesign.	Use for classroom practice that may be revised later.
Shape From Text on a face	Pad or Pocket	Text becomes closed geometry that can add or remove material.	Try both raised and engraved versions.
Open Draft curve	Keep as path / sweep guide	Open curves do not make pad/pocket solids by themselves.	Reserve these for later sweep / loft exercises.

One-sentence takeaway

do I need a quick solid, or do I need an editable PartDesign feature history?

Bridge to the next module

The same decision logic appears again when curves become sweep paths, loft sections, and profile networks for surfaces.

Recap

Quick recap

- **Hermite**: endpoints + tangents.
- **Cardinal**: interpolating cubic with tension and local control.
- **Bézier**: endpoint interpolation, control polygon, de Casteljau.
- **B-spline**: piecewise polynomial, local control, knots.
- **NURBS**: weighted rational extension of B-splines.